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causalCmprsk: An R package for nonparametric and Cox-based estimation of average treatment effects in competing risks data

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Abstract

Background and Objective: Competing risks data arise in both observational and experimental clinical studies with time-to-event outcomes, when each patient might follow one of the multiple mutually exclusive competing paths. Ignoring competing risks in the analysis can result in biased conclusions. In addition, possible confounding bias of the treatment-outcome relationship has to be addressed, when estimating treatment effects from observational data. In order to provide tools for estimation of average treatment effects on time-to-event outcomes in the presence of competing risks, we developed the R package **causalCmprsk**. We illustrate the package functionality in the estimation of effects of a right heart catheterization procedure on discharge and in-hospital death from observational data.

Methods: The **causalCmprsk** package implements an inverse probability weighting estimation approach, aiming to emulate baseline randomization and alleviate possible treatment selection bias. The package allows for different types of weights, representing different target populations.

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Declaration of competing interest

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Supplementary material

Supplementary material associated with this article can be found in the online version.

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causalCmprsk builds on existing methods from survival analysis and adapts them to the causal analysis in non-parametric and semi-parametric frameworks.

Results: The **causalCmprsk** package has two main functions: `fit.cox` assumes a semi-parametric structural Cox proportional hazards model for the counterfactual cause-specific hazards, while `fit.nonpar` does not impose any structural assumptions. In both frameworks, **causalCmprsk** implements estimators of (i) absolute risks for each treatment arm, e.g., cumulative hazards or cumulative incidence functions, and (ii) relative treatment effects, e.g., hazard ratios, or restricted mean time differences. The latter treatment effect measure translates the treatment effect from probability into more intuitive time domain and allows the user to quantify, for example, by how many days or months the treatment accelerates the recovery or postpones illness or death.

Conclusions: The **causalCmprsk** package provides a convenient and useful tool for causal analysis of competing risks data. It allows the user to distinguish between different causes of the end of follow-up and provides several time-varying measures of treatment effects. The package is accompanied by a vignette that contains more details, examples and code, making the package accessible even for non-expert users.

Keywords

Cause-specific hazards; Cumulative incidence functions; Proportional hazards model; Restricted mean time; Right Heart Catheterization data; Time-varying hazard ratio

1 Introduction

Competing risks data are time-to-event data where each individual is followed until the earliest event, out of multiple possible events, occurs. Competing risks arise in numerous real life contexts and are omnipresent in clinical studies with time-to-event outcomes, including randomized controlled trials. For example, in a drug repurposing study for Alzheimer's disease, death might preclude the onset of Alzheimer's disease [1]. Similarly, in a study of the effect of a prior cerebrovascular disease in COVID-19 patients on the time-to-extubation, death might preclude successful extubation [2]. In a study of the effects of a right heart catheterization (RHC) procedure in critically ill patients [3], the stay in a hospital might be terminated due to discharge or due to death (Figure 1) representing quite opposite outcomes that have to be distinguished in the data analysis. Importantly, competing risks cannot be eliminated by a study design and therefore have to be addressed at the stage of data analysis. When the focus is on estimation of treatment effects from observational data where treatments are not randomly assigned, we additionally have to account for possible treatment selection bias.

There are several R packages, developed for analysis of competing risks data, that allow estimation of treatment effects from observational data, in the presence of competing risks. The package **cmprskcoxmsm** can be used for estimation of both the absolute risks, i.e., cumulative incidence functions, and relative treatment effects, such as time-invariant hazard ratios, risk ratios and risk differences for a specific time point. However, **cmprskcoxmsm** relies on the assumed marginal structural Cox model for counterfactual cause-specific hazards [4]. The package **cmprsk** was developed for regression modeling of subdistribution

functions, introduced in [5], and allows for nonparametric estimation. The package **CFC** was developed for Bayesian estimation of cumulative incidence functions in competing risks data using both parametric and nonparametric survival models [6]. However, both **cmprsk** and **CFC** do not have an option to include case weights, thereby not providing the flexibility in balancing treatment arms and adjusting for confounders using weighting approaches. A recently developed package **HazardDiff** can be used for estimation of a conditional treatment effect expressed as time-invariant hazard differences [7]. The package **NPMLEcmprsk** provides tools for nonparametric maximum likelihood estimation of cause-specific risks and hazards as described in [8]. The function `ate` in the package **riskRegression** can be used for estimation of both treatment-specific risks and average treatment effects (risk differences and ratios). The function `ate` implements three estimation approaches, i.e., the g-formula, the inverse probability weighting (IPW), and the doubly robust estimation, as described in [9], but similar to the package **cmprskcoxsm**, `ate` in **riskRegression** relies on the Cox proportional hazards assumption. The **prodlm** package provides the Aalen-Johansen nonparametric estimator [10] of the absolute (but not relative) risks. In particular, the function `prodlm` has an input argument `caseweights`, which can be used to adjust for measured confounders by inverse probability weighting [11]. The package **timereg** provides tools for flexible regression modeling of time-to-event data (see, e.g., [12, 13]) and can be used for estimation of causal effects as well. Recently, the package **survival** [14] was extended from univariate survival models to multi-state models, of which a competing risks model is a special case [15]. The package **survival** includes parametric, semi-parametric [5], and nonparametric [10] estimators of the absolute risks and rates and the covariate effects. There are several available R packages developed for a semi-competing risks model, also known as an illness-death model, where there is an additional transition from a non-terminal ‘discharge’ state to a terminal ‘death’ state. This is a generalization of the competing risks, and it is not considered in this paper.

In this paper, we introduce the R package **causalCmprsk**, which provides a useful complement to the toolbox of existing methods. In **causalCmprsk** all the functionals can be estimated based on nonparametric or semi-parametric proportional hazards Cox-based approaches, and the proportional hazards assumption can be visually tested. **causalCmprsk** implements an inverse probability weighting approach and provides a variety of case weights: stabilized and non-stabilized weights for the overall, treated and control populations, and overlap weights for the overlap population, which is the population with the largest equipoise regarding the treatment. In **causalCmprsk** we implemented weighted bootstrap for quantification of uncertainty that accounts for the sampling variability of estimated weights, an issue that is often overlooked in practice [16]. In **causalCmprsk**, there is an option of parallel computing to speed up bootstrap computations. **causalCmprsk** provides several measures of treatment effects on time-to-event outcomes, among which the restricted-mean-time differences are the most appealing - they are able to provide an intuitive summary of treatment effects and quantify how much time on average was lost (or gained) due to treatment. We illustrate the functionality of **causalCmprsk** by analyzing a real data set, described in Section 2. In Section 3, we provide the methodological background and details on the estimation approaches implemented in **causalCmprsk**. Section 4 illustrates the package functionality through the comprehensive

analysis of the RHC data. The summary of the capabilities of **causalCmprsk** and its limitations are discussed in Section 5.

2 Right Heart Catheterization data

We will illustrate how to use **causalCmprsk** by analyzing data from an observational study on a RHC procedure. The data were obtained from <http://hbiostat.org/data/>. The RHC data were collected by Connors et al. [3] with the aim to evaluate the effectiveness and safety of the RHC procedure for critically ill patients. The study of Connors et al. [3] appeared to be influential since most of the physicians strongly believed in the benefits of the RHC procedure, and therefore conducting a clinical trial to test RHC was unethical. Connors et al. collected an extensive set of covariates in order to correct for possible treatment assignment bias, and their results showed that RHC might be harmful. Following that, these data drew attention of many researchers. The data were reanalyzed multiple times, for example, by [17, 18, 19, 20, 21, 22, 23, 24], among many others. All of these studies estimated the effect of RHC on mortality and consistently showed that undergoing RHC leads to higher mortality rate and risk.

This RHC data set contains information on 5735 critically ill patients who were considered for a RHC procedure upon their admission to a hospital. Out of 5735 patients, 2184 underwent a RHC procedure and the remaining 3551 did not. The information on their baseline characteristics includes 65 covariates.

The data were previously analyzed for different outcomes derived from the original data set. Initially, the interest was in 30-day survival, which was summarized with a binary indicator of 30-day survival, used as an outcome by [3, 19, 21, 23] and [24]. Tan (2006) [18] used a survival outcome with censoring at day 30. Mao and Li (2020) [20] and Keele and Small (2021) [22] analyzed data for the length-of-stay in a hospital, without distinguishing whether the end of stay was due to discharge or due to in-hospital death, combining the two opposite causes of the end of stay into one composite outcome. If a discharge from a hospital can be thought as a measure of clinical improvement, then combining these two causes for the end of stay into one composite endpoint might be problematic.

In this paper, we reanalyze the RHC data for the length-of-stay accounting for the cause of the end of stay. Using the R package **causalCmprsk**, we estimate the effects of RHC on two outcomes, time-to-discharge and time-to-in-hospital-death. The competing risks analysis in Sections 4.2 and 4.3 provides better understanding of the effects of the RHC procedure on the overall hospital length-of-stay. We compare our results to the findings of [20]. We also analyze the RHC data for 30-day survival in Section 4.4. These results then are compared to the results of the original study of Connors et. [3].

3 Competing risks framework

3.1 Notation and assumptions

Without loss of generality, in what follows we introduce notation using the RHC data example, where everyone starts at admission to an intensive care unit (ICU), and there are

two types of events that can terminate the follow-up (Figure 1), though **causalCmprsk** can be used for analysis of data with more than two competing events.

We suppose that the observed data comprise n independent observations of $(T, E, A, \mathbf{C})$, i.e., $(t_i, e_i, a_i, \mathbf{c}_i)$, for $i = 1, \dots, n$. Let T denote the time between the **Admission to an ICU** and the **Discharge** or **In-hospital death**, whichever comes first. Let E denote the type of event, with $E = 1$ if T corresponds to the time-to-discharge, and $E = 2$ if T corresponds to the time-to-in-hospital-death. If none of the two events of interest are observed during the follow-up, T is the time-to-censoring, and in this case $E = 0$. In the RHC data, the date of discharge (either alive or due to death) is known for all patients except for one person. Let $A \in \{0, 1\}$ denote the time-invariant treatment random variable, with $A = 1$ for those who underwent the RHC procedure at the time of admission to an ICU, and $A = 0$ for those who did not. Let $\mathbf{C}$ be a vector of baseline covariates.

We use the potential outcomes framework [25]. Let (T^a, E^a) denote the potential outcomes that would have been observed had the treatment been set to $A = a$, for $a = 0, 1$. Throughout, we assume the following causal assumptions:

CA1. No unmeasured confounding: Treatment A is independent of potential outcomes given $\mathbf{C}$. That is, given $\mathbf{C}$, A and (T^a, E^a) are independent, for $a = 0, 1$;

CA2. Positivity (overlap): $0 < P(A = a | \mathbf{C}) < 1$ for $a = 0, 1$;

CA3. Stable unit treatment value assumption: The outcome of every patient does not depend on the treatment of others (noninterference), and the outcome does not depend on the way a treatment was assigned (consistency). The observed outcomes (T, E) are related to potential outcomes (T^a, E^a) , $a = 0, 1$, by $T = AT^1 + (1 - A)T^0$ and $E = AE^1 + (1 - A)E^0$.

In the presence of right censoring, we additionally assume that the potential right censoring time is independent of the potential outcomes (T^a, E^a) and conditionally independent of the treatment variable A given covariates $\mathbf{C}$.

3.2 Treatment effect estimands

In order to introduce treatment effect estimands, we first define single-world cause-specific hazards of transitioning to states **1. Discharge** or **2. In-hospital death** (Figure 1) in a counterfactual world where everyone would receive treatment a

$$h_k^a(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(t \leq T^a < t + \Delta t, E^a = k | T^a \geq t), \quad k = 1, 2,$$

and single-world cumulative incidence functions (CIF)

$$CIF_a(t, k) = E[I(T^a \leq t, E^a = k)] = P(T^a \leq t, E^a = k) = \int_0^t h_k^a(s) S_a(s) ds, \quad k = 1, 2,$$

where $I(\cdot)$ is the indicator function that equals 1 if the condition inside the parentheses is true and 0 otherwise, and $S_a(t) = P(T^a \geq t)$ is an overall survival function in the counterfactual world where everyone would receive treatment a . Hazards $h_k^a(t)$ represent (possibly time-varying) *rates* of events, while $CIF_a(t, k)$ are often interpreted as *risks* of events, see, e.g., [26, 27].

The following restricted mean time (RMT) functionals

$$RMT_a(\tau, k) = \int_0^\tau CIF_a(t, k) dt, \quad k = 1, 2, \quad a = 0, 1,$$

can be used to transform $CIF_a(t, k)$ from probabilities to the time domain. In the above, τ is some prespecified time point of interest, e.g., 5 years. $RMT_a(\tau, k)$ is the area under the $CIF_a(t, k)$ curve. If the events are desirable, we would like to see them happen faster, meaning that the $CIF_a(t, k)$ curve will increase fast and steeply with t , resulting in a large area under the curve, i.e., larger $RMT_a(\tau, k)$. In addition, since the maximum value of $RMT_a(\tau, k)$ we could have is τ , for shorter time-to-events T (or quicker transitions to a desirable state) we will have more time to spend in that state, since shorter T will lead to longer $\tau - T$. For example, we would like for patients to recover faster, which would result in longer average time in recovery, see, e.g., [28].

Hazards ratios (HR), risk differences (RD) and risk ratios (RR).—The hazard ratios for both events are defined by

$$HR_k(t) = \frac{h_k^1(t)}{h_k^0(t)}, \quad k = 1, 2. \quad (1)$$

Notice that $HR_k(t)$ in (1) can depend on t , since we do not make any modeling assumptions regarding the hazard functions $h_k^a(t)$.

The risk differences are defined by

$$RD(t, k) = E\left[I\left(T^1 \leq t, E^1 = k\right)\right] - E\left[I\left(T^0 \leq t, E^0 = k\right)\right] = CIF_1(t, k) - CIF_0(t, k), \quad k = 1, 2.$$

$RD(t, k)$ are between-treatment-arms differences of probabilities of having the event of type k by time t , before a competing event occurs. They represent the average treatment effects (ATE) on different types of events.

The risk ratios are defined by

$$RR(t, k) = \frac{E\left[I\left(T^1 \leq t, E^1 = k\right)\right]}{E\left[I\left(T^0 \leq t, E^0 = k\right)\right]} = \frac{CIF_1(t, k)}{CIF_0(t, k)}, \quad k = 1, 2.$$

Restricted mean time (RMT) gained/lost due to treatment.—If we want to quantify the effect of treatment in terms of the number of days/months/years gained (or lost) on average from being treated, we can define the ATE based on RMT as the difference

$$RMT_1(\tau, k) - RMT_0(\tau, k), \quad k = 1, 2. \quad (2)$$

The use of the RMT difference for quantifying treatment effects is becoming more popular in the univariate survival analysis, where this effect is usually defined as the difference between the restricted mean survival times given by

$$RMST_a(\tau) = E\{T^a I(T^a \leq \tau)\},$$

which is the expectation of the restricted time-to-event, where an “event” in this case is a composite outcome, that is any event. However, in competing risks, the cause-specific RMT differences (2) are still rarely used. Several recent papers, e.g., [28, 29, 30], have real world data examples illustrating the use of RMT-based differences for quantification of average treatment effects in the competing risks setting.

From the definition of $RMST_a(\tau)$, it is clear that this quantity does not distinguish between the types of events, but only uses the information on the times to events. In competing risks, $RMST_a(\tau)$ can be interpreted as the expected restricted sojourn time in state 0 (Figure 1). The relation between $RMST_a(\tau)$, used in the univariate survival analysis, and the cause-specific $RMT_a(\tau, k)$, $k = 1, 2$, used in the competing risks analysis, can be derived as follows

$$\begin{aligned} RMST_a(\tau) &= E\{T^a I(T^a \leq \tau)\} = \int_0^\tau \{1 - CIF_a(t, 1) - CIF_a(t, 2)\} dt \\ &= \tau - \int_0^\tau CIF_a(t, 1) dt - \int_0^\tau CIF_a(t, 2) dt = \tau - RMT_a(\tau, 1) - RMT_a(\tau, 2). \end{aligned}$$

That is, $RMT_a(\tau, 1)$ and $RMT_a(\tau, 2)$, combined together, complement $RMST_a(\tau)$.

3.3 Estimation approaches implemented in causalCmprsk

3.3.1 Inverse probability weighting using propensity scores—When estimating ATEs from observational data, where the treatment is not randomized, we have to account for possible treatment selection bias. Under causal assumptions listed in Section 3.1, the ATEs are identifiable and can be estimated. The **causalCmprsk** package implements an IPW estimation approach [11], based on a treatment assignment model, also known as a propensity scores (PS) model. A propensity score for an individual with observed vector of covariates $\mathbf{C} = \mathbf{c}_i$ is defined as $ps_i = P(A = 1 | \mathbf{C} = \mathbf{c}_i)$ [31]. Further, we denote the estimates of ps_i by $\widehat{ps}_i$.

The process of building a PS model involves several critical steps that have to be taken into account when using an IPW estimation approach: choosing a PS model, selecting variables to include in this model, choosing appropriate balancing weights based on a fitted PS model,

employing appropriate PS model diagnostics, and finally, taking into account the uncertainty resulting from a treatment assignment modeling step. The aim of the PS model is to balance covariates between treatment groups, and there are several proposed ways to achieve this balance. The **causalCmprsk** package takes a step-by-step approach, where any part in the process can be replaced with an alternative, e.g., another type of a PS model, or another type of weights. In what follows, we discuss some of the aspects of building a PS model:

- A.** A fitted PS model can be formally assessed for the goodness-of-fit (GOF) to the observed data using formal statistical tests. However, this type of diagnostics is usually not done in practice. The reason is that the primary goal of the PS model is to balance baseline covariates between treatment arms, therefore the main criterion of a goodness-of-fit for the PS model is usually the degree of the achieved balance for important confounders. **causalCmprsk** uses logistic regression for the propensity scores model. All the covariates provided by the user to the functions `get.weights`, `fit.nonpar` and `fit.cox` are included in the PS model, and no variable selection is implemented. Both original covariates and their interactions can be included in the PS model. Provided a PS model fits the data well, and in the absence of non-overlap (see the assumption CA2), the covariates in the weighted population will be balanced, meaning that the weighted marginal distributions will be similar in two treatment arms (if their main effects were included in the PS model), and the weighted joint distributions will be similar in both treatment arms as well (if their corresponding interactions were included in the PS model). The problem of non-overlap can be overcome, e.g., by using overlap weights [19, 32, 33].
- B.** For theoretical and practical considerations on confounder selection we refer to [34, 35, 36]. There are some general results, e.g., the backdoor and front-door criteria, for selection of adjustment variables using Directed Acyclic Graphs (DAGs). Currently, **causalCmprsk** does not implement any confounder selection procedure, and all the factors provided by the user to the functions `get.weights`, `fit.nonpar` and `fit.cox` in the argument `C` are included in the PS model. In general, the process of variable selection in the PS model involves not only a covariate-treatment relationship but also a covariate-outcome relationship, since only variables that are related to both the treatment and the outcome are defined as confounders. It is recommended that the list of potential confounders be screened by subject matter experts to avoid inclusion of possible instrumental variables (IV) in the PS model, since this might amplify the residual bias in the presence of unmeasured confounding or increase the variance if all the confounders are measured [34].
- C.** It is important to diagnose a fitted PS model before we use it, since it is the basis for our inference on the average treatment effects. The standard statistical diagnostic test for the GOF of a logistic regression model, the Hosmer-Lemeshow test [37], is very rarely used for diagnostics of a PS model [38]. Although it is important that the model-based predicted probabilities of treatment assignment are similar to the corresponding empirical probabilities [19], most often a fitted PS model is assessed by its ability to balance baseline covariates,

which is verified by looking at the absolute standardized differences [19, 33, 35, 38]. For continuous covariates, it is even more informative to compare the whole distributions of covariates between treatment groups, and not only their weighted and standardized averages. Moreover, given the same PS model, the attained covariate balance depends on the type of weights we use.

In order to diagnose a fitted PS model and check the resulting covariate balance, we recommend, first, to call function `get.weights`, which can be done prior to any estimation of ATEs. We illustrate how to verify visually whether the PS model fits the data well, and to evaluate the attained covariate balance in the vignette accompanying **causalCmprsk**.

- D.** The choice of weights plays a crucial role in the estimation of average treatment effects, since weights define the composition of the target population for which we estimate the ATE. If treatments have heterogeneous effects on patients, different weighting schemes might lead to quite different results, because of the different compositions of the target population [39].

The package **causalCmprsk** has the following types of weights (`wtype` argument in functions `fit.nonpar`, `fit.cox` and `get.weights`), which correspond to different target populations:

1. "ATE" uses $\hat{w}_i = \frac{1}{\hat{P}(A = a_i | C = c_i)}$, which corresponds to the ATE for the overall cohort;
2. "stab.ATE" uses $\hat{w}_i = \frac{\hat{P}(A = a_i)}{\hat{P}(A = a_i | C = c_i)}$, which is a stabilized version of ATE weights [40], so that "stab.ATE" corresponds to the ATE for the overall cohort;
3. "ATT" uses $\hat{w}_i = \frac{\hat{P}(A = 1 | C = c_i)}{\hat{P}(A = a_i | C = c_i)} = \frac{\hat{p}s_i}{\hat{P}(A = a_i | C = c_i)}$, which corresponds to the ATE for the population of treated patients;
4. "ATC" uses $\hat{w}_i = \frac{\hat{P}(A = 0 | C = c_i)}{\hat{P}(A = a_i | C = c_i)} = \frac{1 - \hat{p}s_i}{\hat{P}(A = a_i | C = c_i)}$, which corresponds to the ATE for the population of untreated patients (i.e. a control group);
5. "overlap" uses $\hat{w}_i = \frac{\hat{P}(A = 1 | C = c_i)\hat{P}(A = 0 | C = c_i)}{\hat{P}(A = a_i | C = c_i)} = \frac{\hat{p}s_i(1 - \hat{p}s_i)}{\hat{P}(A = a_i | C = c_i)}$, which corresponds to the ATE for the overlap population. Please see Table 1 in [19].

Since the different weighting schemes target different population average effects, it is important to characterize this target population [41], e.g., by describing covariate distributions of this target population. We illustrate this step in the vignette accompanying **causalCmprsk**.

3.3.2 Cox-based estimation of average treatment effects—Function `fit.cox` implements the Cox-based estimation by assuming the Cox proportional hazards model for the single-world cause-specific hazards

$$h_k^0(t) = h_{0k}(t); \quad h_k^1(t) = h_{0k}(t)\exp(\beta_k), \quad k = 1, 2. \quad (3)$$

The model (3) assumes that the cause-specific hazards are proportional in two counterfactual worlds, a world where everyone would be treated and a world where everyone would be untreated, and a proportionality factor between them is $\exp(\beta_k)$ and $HR_k(t) = HR_k$. That is, the outcome model (3) implies time-invariant hazard ratios HR_k . For more details on the implemented Cox-based estimation, and for the explicit expressions of the estimating equations and the estimators we refer to [1].

3.3.3 Nonparametric estimation of average treatment effects—Function `fit.nonpar` implements nonparametric estimation. In nonparametric survival framework, the single-world cause-specific hazards $h_k^0(t)$ and $h_k^1(t)$ ($k = 1, 2$) stay unspecified, that is $h_k^0(t)$ and $h_k^1(t)$ vary freely and are not tied together by any assumptions. This implies that $HR_k(t)$ in (1) can be time-dependent. The nonparametric estimators provide a way to visually test the proportional hazards assumption. We illustrate how to do this in the RHC data analysis in Sections 4.2 and 4.4.

The nonparametric estimation procedure includes, first, estimation of all single-world cumulative cause-specific hazard functions, $H_k^a(t) = \int_0^t h_k^a(s)ds$, $k = 1, 2$, $a = 0, 1$. For the estimation of $H_k^a(t)$, **causalCmprsk** uses a weighted version of the Nelson-Aalen estimator:

$$\hat{H}_k^a(t) = \sum_{i: t_i \leq t, a_i = a, e_i = k} \frac{\sum_{j: a_j = a, e_j = k} \hat{w}_j I(t_j = t_i)}{\sum_{j: a_j = a} \hat{w}_j I(t_i \leq t_j)}, \quad k = 1, 2, \quad a = 0, 1.$$

The balancing weights $\hat{w}_i$ that aim to adjust for the confounding bias, are time-invariant subject-specific weights, where the contribution of the i^{th} observation is $\hat{w}_i$ instead of 1. Then, using $H_k^a(t)$ as building blocks, the estimators of all other parameters are derived by plugging-in the estimated hazards $\hat{H}_k^a(t)$:

- The estimator of $CIF_a(t, k)$ is given by

$$\widehat{CIF}_a(t, k) = \frac{\sum_{i: t_i \leq t, a_i = a, e_i = k} \hat{P}_a(T > t_i) - \frac{\sum_{j: a_j = a, e_j = k} \hat{w}_j I(t_j = t_i)}{\sum_{j: a_j = a} \hat{w}_j I(t_i \leq t_j)}}{1}, \quad k = 1, 2, \quad a = 0,$$

where $\hat{P}_a(T > t)$ is the weighted Kaplan-Meier estimator of the overall survival function $P_a(T > t)$, i.e., the survival function for the composite time to any event, in the treatment arm a . The estimator $\widehat{CIF}_a(t, k)$ is the weighted Aalen-Johansen estimator of the CIFs.

- The estimator of $RMT_a(t, k)$ is

$$\widehat{RMT}_a(\tau, k) = \int_0^\tau \widehat{CIF}_a(t, k) dt, \quad k = 1, 2, \quad a = 0, 1.$$

- The estimators of the treatment effect measures, i.e., $HR_k(t)$, $RD(t, k)$, $RR(t, k)$ and $RMT_1(\tau, k) - RMT_0(\tau, k)$, are obtained by plugging $\widehat{H}_k^1(t)$ and $\widehat{H}_k^0(t)$ into the respective functional.

Finally, we notice that in both Cox-based and nonparametric estimation frameworks, all the resulting estimators, except for the estimators of $RMT_a(\tau, k)$ and their differences, are *càdlàg* step functions, while the estimators of $RMT_a(\tau, k)$ and $RMT_1(\tau, k) - RMT_0(\tau, k)$ are continuous and smooth functions.

3.3.4 Quantification of uncertainty—causalCmprsk implements Bayesian bootstrap ([42, 43]), which is also called weighted bootstrap to obtain pointwise confidence intervals (see, e.g., [44], pp. 179–180 in [45], [46]). In survival data, the simple bootstrap can be problematic because it might lead to a large proportion of tied survival times, and in the presence of censoring, it is possible that a bootstrap sample is comprised of only censored observations ([45, 46]). To avoid these complications, weighted bootstrap is often used with survival data. In this approach, each observation from the original sample is assigned a random positive weight. The weights w_i^{bs} are constructed by, first, generating random independent numbers $V_i \sim \text{Exp}(1)$, $i = 1, \dots, n$, then truncating them at 5, $W_i = \min(V_i, 5)$, and lastly standardizing W_i so that the sum of weights of all observations would be equal to the sample size, i.e., $w_i^{bs} = \frac{W_i}{\bar{W}}$, where $\bar{W} = \frac{1}{n} \sum_{i=1}^n W_i$. Truncation at 5 is used to prevent from observations with rarely sampled extremely high weights (the probability of this is 0.006738) to dominate a bootstrap sample. The described procedure is equivalent to assigning weights from *Dirichlet*(1, ..., 1) distribution.

For each bootstrap sample, we repeat the steps of: (a) fitting the propensity scores model and obtaining balancing weights, and (b) estimating all the parameters of interest. Importantly, this bootstrap estimator refits a propensity score model for each bootstrap sample, which results in variance estimators that naturally account for the sampling variability in the estimated subject-specific weights, an issue that is often overlooked in statistical analyses with propensity scores weighting. All our parameters, except for β_k under the model (3), are time-dependent, and their pointwise 95% confidence intervals are obtained as 0.025 and 0.975 quantiles from the sample distributions of bootstrap estimates for each time point separately.

4. Analysis of RHC data using causalCmprsk

4.1 Data preparation

The R code for loading, preparing the RHC data for the analysis using the **causalCmprsk** package, and all the analyses presented in this paper is provided in the Supplementary Appendix associated with this article, which can be found in the online

version, and also in the vignette accompanying **causalCmprsk** at https://cran.r-project.org/package=causalCmprsk/vignettes/cmp_rsk_RHC.html.

The required data format is one row per person. The following 8 observations from the RHC data, loaded into an R data.frame object named `rhc_full`, illustrate the data format required for **causalCmprsk**:

```
> rhc_full[1:8,c("Time", "E", "RHC", "T.death", "D", "T.death.30", "D.30",
+               "age", "sex_Male")]
```

	Time	E	RHC	T.death	D	T.death.30	D.30	age	sex_Male
1	9	1	0	240	0	30	0	70.25098	1
2	45	2	1	45	1	30	0	78.17896	0
3	60	1	1	317	0	30	0	46.09198	0
4	37	2	0	37	1	30	0	75.33197	0
5	2	2	1	2	1	2	1	67.90997	1
6	7	1	0	201	0	30	0	86.07794	0
7	42	1	0	235	0	30	0	54.96799	1
8	34	1	0	38	1	30	0	43.63898	1

The data must include a time-to-event variable, a categorical indicator of the event type or censoring and a binary treatment indicator, e.g., for the competing risks analysis of the RHC data described in Section 4.2, this triplet is (Time, E, RHC).

In the RHC data, out of 5735 individuals, there were 3704 patients who were discharged alive (E=1), and 2030 patients who died while staying in a hospital (E=2):

```
> E <- as.factor(rhc_full$E)
> levels(E) <- c("Discharge", "In-hospital death")
> addmargins(table(E, useNA="ifany"))
E
```

Discharge	In-hospital death	<NA>	Sum
3704	2030	1	5735

In our competing risks analysis in Section 4.2, the variable Time denotes the length of stay in a hospital, and the variable E indicates the type of event that terminated the hospitalization (**1.Discharge alive** or **2.In-hospital death**). There is one observation with a missing date of discharge, but with a known date of death. Therefore, we will exclude it from the competing risks analysis of Section 4.2 and 4.3, but will include it in the analysis of 30-day survival of Section 4.4. For those patients who were discharged alive, the follow up continued after their discharge. Overall, 3722 patients died by the end of the study:

```
> D <- as.factor(rhc_full$D)
> levels(D) <- c("censored", "died")
> addmargins(table(D, useNA="no"))
D
censored    died    Sum
2013       3722    5735
```

There were 1918 patients who died during the first 30 days after admission to an ICU (D.30=1), and 3817 subjects whose time-to-death is right-censored (D.30=0) in the analysis of 30-day survival:

```
> D.30 <- as.factor(rhc_full$D.30)
> levels(D.30) <- c("censored", "died")
> addmargins(table(D.30, useNA="no"))
D.30
```

censored	died	Sum
3817	1918	5735

4.2 Estimation of RHC effects on discharge and in-hospital death

In this competing risks analysis, we aim to distinguish between two types of events that could terminate the hospitalization, **Discharge alive** and **In-hospital death**, and also to allow for treatment effects to vary over time. Both these goals can be easily achieved with the **causalCmprsk** package. We will illustrate how to estimate all the parameters defined in Section 3.2, using both nonparametric and Cox-based approaches.

Two main functions of **causalCmprsk**, `fit.cox` and `fit.nonpar`, fit a propensity score model internally, and there is no need in estimation of propensity scores prior to calling `fit.cox` or `fit.nonpar`. The argument `trt.formula` is a formula object of R, with a treatment indicator as a response variable appearing on the left hand side of a `trt.formula` and an arithmetic expression involving all the confounders appearing on its right hand side. Since our analysis uses 65 confounders available in the RHC data, we do not include the R code that creates `trt.formula` in the main text of this paper, but it is provided in the Supplementary Appendix associated with this article, which can be found in the online version, and also it is available in the vignette accompanying **causalCmprsk** at https://cran.r-project.org/package=causalCmprsk/vignettes/cmp_rsk_RHC.html

```
library(causalCmprsk)
res.nonpar <- fit.nonpar(df=rhc, X="Time", E="E", trt.formula=trt.formula,
                        wtype="overlap", cens=0, conf.level=0.95,
                        bs=TRUE, nbs.rep=200, seed=17, parallel = TRUE)
```

```
res.cox <- fit.cox(df=rhc, X="Time", E="E", trt.formula=trt.formula,
                  wtype="overlap", cens=0, conf.level=0.95, bs=TRUE,
                  nbs.rep=200, seed=17, parallel = TRUE)
```

We did not assess the confounder selection via DAGs, but instead built on the decisions made by the original study's investigators who consulted the subject matter experts when collected the RHC data [3]. Our analysis is based on overlap weights. There are several available R packages developed for diagnostics of weights and assessing balance between treatment groups. For a visual inspection of covariate balance between treatment groups, in the analysis of the RHC data we looked at the absolute standardized differences using `love.plot()` function from the **cobalt** package [47]. The R code for the balance diagnostics and visualization for the RHC data is available in the vignette accompanying **causalCmprsk**.

Both `fit.cox` and `fit.nonpar` have the same 14 input arguments described in Table 1. Functions `fit.cox` and `fit.nonpar` use bootstrapping for estimation of standard errors and confidence intervals of the estimated parameters. If the argument `parallel` is set to `TRUE`, the bootstrap resampling and estimation is conducted in parallel. The number of available cores is detected automatically, and the parallel jobs are assigned to the number of detected available cores minus one. In the above application with 200 bootstrap replications, the average execution time of 50 runs of `fit.nonpar` on the RHC data was 27 seconds, and for `fit.cox` it was 55 seconds. In contrast, running `fit.nonpar` and `fit.cox` with 200 sequentially executed bootstrap replications resulted in average execution time of 75 and 163 seconds, respectively, which is almost 3 times slower compared to the parallel execution. This experiment was conducted on Intel(R) Core(TM) i7-8565U CPU 1.80 GHz with 32.00 GB RAM, and the number of available cores, detected by `detectCores` function from the R package **parallel**, was 8.

Graphical test of a proportional hazards assumption—A standard graphical way to test a proportional hazards assumption is by plotting the log-ratios, $\log \frac{H_k^1(t)}{H_k^0(t)}$, $k = 1, 2$, versus t , with $H_k^a(t)$, $a = 0, 1$, estimated nonparametrically. If the proportional hazards model (3) holds, these log-ratios are expected to be time-independent. The R code below illustrates how to create such a plot, and Figure 2 shows the results.

```
library(ggplot2)
library(dplyr)
library(ggalt)
df1 <- rbind(cbind(summary(res.nonpar, event=1, estimand="logHR") %>%
                    select(time, logCumHazR, CIL.logCumHazR, CIU.logCumHazR),
                    Outcome.method=1),
             cbind(summary(res.nonpar, event=2, estimand="logHR") %>%
                    select(time, logCumHazR, CIL.logCumHazR, CIU.logCumHazR),
                    Outcome.method=2))
HR1 <- res.cox$trt.eff[[paste("Ev=", 1, sep="")]][c("log.CumHazR",
```

```

      "log.CumHazR.CI.L", "log.CumHazR.CI.U"]
HR2 <- res.cox$trt.eff[[paste("Ev=", 2, sep="")]][c("log.CumHazR",
      "log.CumHazR.CI.L", "log.CumHazR.CI.U")]
df2 <- data.frame(time=c(min(res.cox$time), max(res.cox$time)),
      logCumHazR=HR1[[1]], CIL.logCumHazR=NA,
      CIU.logCumHazR=NA, Outcome.method=3)
df3 <- data.frame(time=c(min(res.cox$time), max(res.cox$time)),
      logCumHazR=HR2[[1]], CIL.logCumHazR=NA,
      CIU.logCumHazR=NA, Outcome.method=4)
df <- rbind(df1, df2, df3)

df$Outcome.method <- factor(df$Outcome.method)
levels(df$Outcome.method) <- c("Discharge-nonpar", "Death-nonpar",
      paste("Discharge-Cox:", round(HR1[[1]], dig=2),
      ", 95% CI=[" , round(HR1[[2]], dig=2),
      ",", round(HR1[[3]], dig=2), "]", sep=""),
      paste("Death-Cox:", round(HR2[[1]], dig=2),
      ", 95% CI=[" , round(HR2[[2]], dig=2),
      ",", round(HR2[[3]], dig=2), "]", sep=""))

ggplot(df, aes(x=time, y=logCumHazR, color=Outcome.method,
      fill=Outcome.method, shape=Outcome.method)) +
  geom_step(linewidth=1.1) +
  geom_ribbon(aes(ymin=CIL.logCumHazR, ymax=CIU.logCumHazR),
      alpha=0.2, stat="stepribbon") +
  scale_color_brewer(palette = "Set1") +
  scale_fill_brewer(palette = "Set1") +
  xlab("time from admission to ICU (days)") + ylab("log.CumHR(t)") +
  theme(legend.position = c(0.58, 0.3), legend.title=element_blank(),
      legend.text=element_text(size=12),
      legend.background=element_rect(fill="transparent")) +
  geom_vline(xintercept=30, linetype="dashed")

```

From Figure 2, it is clear that both hazard ratios $HR_k(t)$, $k = 1, 2$, are not time-invariant. The effect of RHC on the *discharge rate*, measured by the hazard ratio $HR_1(t)$, is below 1 early in time (at < 30 days). Later on, the effect of RHC on the discharge rate is gradually decreasing in magnitude until it becomes insignificant. At around 70 days, the effect of RHC on the discharge rate vanishes. The average effect of RHC on the rate of discharges over the whole time period, summarized by the time-fixed hazard ratio HR_1 , obtained from the Cox-based analysis, is significantly lower than 1 (equivalently, $\log HR_1 = \beta_1 < 0$), meaning that, in overall, the RHC reduces the discharge rate.

The rate of *in-hospital deaths* early in time is higher for the RHC treatment. Later on, in the interval between 50 and 150 days after hospitalization, the effect of RHC on the rate of deaths, for those who are still in an ICU, vanishes. Even later, the effect of RHC on the rate

of deaths among those who are in an ICU by day 150 or later, is significantly protective, and those who were treated with RHC die at significantly lower rate than those who did not undergo the RHC procedure. That is, if a patient is still in a hospital at day 150 (i.e., a long-term survivor), and had a RHC procedure, the instantaneous probability of death is much lower than it would be had a patient not be treated with RHC. However, this might be observed due to selection bias, since at day 150 only those patients who survived and stayed in an ICU are included in the analysis. Interestingly, the time-fixed hazard ratio HR_2 shows a significantly harmful effect of RHC on an in-hospital death rate. This time-fixed parameter represents an average rate over the whole time period, and is mostly dominated by the trends in $HR_k(t)$ in the early times, when most of the events happened.

In summary, we conclude that: (i) the proportional hazards assumption does not hold, therefore we will do the rest of the analyses using a nonparametric estimation approach, and (ii) it is important to distinguish between the causes of the end-of-stay in a hospital, since as it can be seen from Figure 2, the rates of two competing events are going in opposite directions, and combining them into one composite outcome might be problematic.

Estimation of cause-specific hazards, CIFs and RMTs.—Relative hazard ratios from Figure 2 do not give the complete picture of treatment effects in the competing risks setting. One of the problems in summarizing treatment effects through hazard ratios is that without a reference hazard function, the hazard ratios do not provide a tool for assessment of full clinical utility of the treatment (see, e.g., [28]). The R code below shows how to extract and plot the estimates of the absolute cause-specific hazards $H_k^a(t)$, which are the building blocks for other parameters, i.e., for $CIF_a(t, k)$ and $RMT_a(t, k)$.

```
df <- rbind(summary(res.nonpar, event=1, estimand="CumHaz"),
            summary(res.nonpar, event=2, estimand="CumHaz"))
df$Event_TRT <- factor(2*(df$Event==2) + 1*(df$TRT==0))
levels(df$Event_TRT) <- c("Discharge-RHC", "Discharge-No RHC",
                        "In-hospital death-RHC", "In-hospital death-No RHC")

ggplot(df, aes(x=time, y=CumHaz, color=Event_TRT, fill=Event_TRT,
               shape=Event_TRT, linetype=Event_TRT)) + geom_step(linewidth=1.1) +
  geom_ribbon(aes(ymin=CIL.CumHaz, ymax=CIU.CumHaz), alpha=0.2,
             stat="stepribbon") + scale_color_brewer(palette = "Set1") +
  scale_fill_brewer(palette = "Set1") +
  xlab("time from admission to ICU (days)") +
  ylab("Cumulative hazard of event by time t") +
  theme(legend.position = c(0.41, 0.83), legend.text=element_text(size=12),
        legend.background=element_rect(fill="transparent"),
        legend.title=element_blank())+
  geom_vline(xintercept=30, linetype="dashed")
```

The results of this analysis are shown on Figure 3 and are summarized in Table 2 for a time point of 30 days. Figure 3(a) shows that, overall, the absolute cumulative hazards of in-hospital deaths are much lower than the cumulative hazards of discharges. Also, it seems that among those who stay in an ICU for longer than 150 days, the absolute cumulative hazard of in-hospital death under No-RHC treatment is higher than under RHC treatment, meaning that RHC treatment seems to be protective for the ICU “long-term survivors”.

Figure 3(b) shows $CIF_a(t, k)$ curves, which were extracted from the object `res.nonpar`, similar to the absolute cumulative hazard functions above. From Figure 3(b), it can be seen that in the presence of in-hospital death, the chances to be eventually discharged alive are lower if a patient undergoes RHC versus not, and the corresponding probabilities are

$$\hat{P}(E^{noRHC} = Discharge) \approx \widehat{CIF}_{noRHC}(180, Discharge) = 0.66, \quad 95\% \text{ CI } [0.64, 0.68],$$

$$\hat{P}(E^{RHC} = Discharge) \approx \widehat{CIF}_{RHC}(180, Discharge) = 0.59, \quad 95\% \text{ CI } [0.57, 0.61].$$

The probability of dying in a hospital is much higher if a patient undergoes RHC versus not:

$$\hat{P}(E^{noRHC} = Death) \approx \widehat{CIF}_{noRHC}(180, Death) = 0.34, \quad 95\% \text{ CI } [0.32, 0.36],$$

$$\hat{P}(E^{RHC} = Death) \approx \widehat{CIF}_{RHC}(180, Death) = 0.41, \quad 95\% \text{ CI } [0.39, 0.43].$$

As an alternative to $CIF_a(t, k)$, we can estimate $RMT_a(t, k)$ by calculating the area under $CIF_a(t, k)$. Figure 3(c) shows the estimates of $RMT_a(t, k)$. Notice that $RMT_a(t, k)$ are not step functions. The interpretation of $RMT_a(t, k)$ is as follows. For a “good” outcome, the larger $RMT_a(t, k)$ is, the better. And, the opposite holds for a “bad” outcome.

The point estimates of $RMT_a(t, k)$ can be easily extracted from `res.nonpar` or `res.cox` objects by running, for example, `get.pointEst(res.nonpar, 30)` for $\tau = 30$ days. This shows the following results: under the RHC treatment, out of 30 days of possible survival, there is an average time loss of 6.8 days (95% CI=[6.4, 7.2]) due to deaths. In contrast, under the No-RHC treatment, the same 30-day average time loss is 5.5 days (95% CI=[5.2, 6.0]), which is a more favorable outcome.

Estimation of various treatment effects of RHC.—In addition to the estimated $HR_k(t)$, representing the effects of RHC on the rates of events, here we illustrate how other treatment effect measures can be estimated and plotted. In the R code below, we create a data frame with RMT differences for both competing events and plot them:

```
df <- rbind(summary(res.nonpar, event=1, estimand="ATE.RMT"),
```

```
summary(res.nonpar, event=2, estimand="ATE.RMT"))
df$Event <- as.factor(df$Event)
levels(df$Event) <- c("Discharge", "In-hospital death")
ggplot(df, aes(x=time, y=ATE.RMT, color=Event, fill=Event,
  shape=Event, linetype=Event)) + geom_line(linewidth=1.1) +
  geom_ribbon(aes(ymin=CIL.ATE.RMT, ymax=CIU.ATE.RMT), alpha=0.2) +
  scale_color_brewer(palette = "Set1") +
  scale_fill_brewer(palette = "Set1") +
  xlab("time from admission to ICU (days)") +
  ylab("average time (days) lost/gained due to treatment") +
  theme(legend.position = c(0.7, 0.5), legend.text=element_text(size=12),
  legend.background=element_rect(fill="transparent"),
  legend.title=element_blank()) +
  geom_vline(xintercept=30, linetype="dashed")
```

Figure 4(a) shows the risk differences $RD(t, k)$ that quantify the treatment effect in terms of probability differences. Thus, for example, we conclude that by day 30, there is 6.7% more chance of in-hospital death (95% CI=[4.1%, 8.8%]) under RHC treatment compared to No-RHC, and 11.0% lower chance to be discharged (95% CI=[8.3%, 13.7%]) under RHC compared to No-RHC. Similar results were obtained for $RR(t, k)$ in terms of probability ratios (Figure 4(b) and Table 2). Figure 4(c) shows $RMT_{RHC}(t, k) - RMT_{No-RHC}(t, k)$ that quantify the size of the treatment effect in terms of expected time lost or gained due to treatment by time t . Thus, for example, we conclude that by day 30, the RHC treatment accelerates death, on average, by 1.29 days (95% CI=[0.78, 1.73]), and postpones discharge, on average, by 2.47 days (95% CI=[1.98, 2.95]). Such “translation” of risk curves into the average time lost due to treatment can be very helpful in understanding the real-life meaning of the effect size and can assist in planning and clinical decision making.

The code that generated all the results presented in this Section, on Figures 3 and 4 and in Table 2 is provided in the Supplementary Appendix associated with this paper. In Sections 4.3 and 4.4 we illustrate how the results obtained from the **causalCmprsk** package can be compared to previous studies that analyzed the RHC data set.

4.3 Estimation of RHC effect on the length-of-stay

Here we compare our results to the findings of Mao and Li [20], who estimated the ATE of RHC on the logarithm of the length-of-stay (LOS) in a hospital, using a varying coefficient regression that allows for nonlinear association between an outcome and a propensity score. Their estimate of the ATE on the log(LOS), based on overlap weights (see their Table 4), is 0.094 (pvalue< 0,001), meaning that RHC significantly increases log(LOS) by 0.094 units (or LOS by $\exp(0,094) = 1,1$ days). LOS is a composite outcome that measures the time spent in a hospital until either death or discharge alive. **causalCmprsk** allows for analysis of univariate survival outcomes as well. For example, the code below conducts the analysis with the univariate LOS outcome:

```

rhc$E.LOS <- rhc$E %in% c(1,2) + 0
res.LOS <- fit.nonpar(df=rhc, X="Time", E="E.LOS", trt.formula=trt.formula,
wtype="overlap", cens=0, conf.level=0.95, bs=TRUE, nbs.rep=200,
seed=17, parallel = FALSE)

```

Figure 5 shows the estimates of the cumulative distribution functions (CDF) for the LOS outcome, i.e., $P(LOS^{RHC} \leq t)$ and $P(LOS^{noRHC} \leq t)$. From Figure 5 and Table 3, we can see that $\hat{P}(LOS^{RHC} \leq 30) = 0.757 < 0.8 = \hat{P}(LOS^{noRHC} \leq 30)$. Or, alternatively,

$$\hat{P}(LOS^{RHC} \geq 30) = 0.243 > 0.2 = \hat{P}(LOS^{noRHC} \geq 30),$$

meaning that under RHC there is higher probability of staying in a hospital longer than without RHC, which is consistent with the findings of both Mao and Li [20] and of the original study of Connors et al. [3]. However, the use of the composite LOS outcome and its application might be problematic in practice. We have to remember that the LOS combines two processes of **Discharge** and **In-hospital death** into one outcome. Although, from Figure 5 and Table 3 we see that the probability of leaving a hospital (either alive or not) by 30 days is estimated significantly lower for the RHC treatment than for the non-RHC, this happens mostly due to the slowing-down effect of RHC on the discharge rate, as can be seen from Figure 3(a). In other words, longer stay in the hospital attributed to the RHC treatment is a result of a slower rate of discharges in that group, and not of longer survival. This example illustrates the importance of distinguishing between two competing processes which might act in the opposite directions, so that combining them into one outcome can be problematic.

4.4 Estimation of RHC effect on 30-day survival

The discharge from a hospital does not necessarily prevent short-term death. Some of the quickly discharged critically ill patients could die soon after their discharge, and therefore their outcome should be counted as a failure and not as a success. For example, an individual 8 in the R output at the end of Section 4.1 died 4 days after his discharge. The RHC data set has the information on the time of death for those who died after the discharge, and we can use **causalCmprsk** for estimation of the RHC effect on 30-day survival, an outcome that is usually of great interest in the critical care medicine.

For this purpose, we define a single time-to-death outcome T.death.30, regardless of whether death occurred in or out of a hospital. For this analysis, all times, which are longer than 30 days, are censored:

```

rhc_full$T.death.30 <- pmin(30, rhc_full$T.death)
rhc_full$D.30 <- rhc_full$D*ifelse(rhc_full$T.death <=30, 1, 0)

```

The nonparametric and Cox-based analyses can be conducted in a similar way as in the competing risks analysis of Section 4.2, but with a single time-to-death outcome T.death.30,

censored at 30 days, and a censoring indicator D.30. For example, the nonparametric analysis can be run by:

```
res.30 <- fit.nonpar(df=rhc_full, X="T.death.30", E="D.30",
                    trt.formula=trt.formula, wtype="overlap", cens=0,
                    conf.level=0.95,
                    bs=TRUE, nbs.rep=200, seed=17, parallel = TRUE)
```

The results are shown on Figure 6 and summarized in Table 3 for a time point of 30 days. Figure 6(a) shows the log of the cumulative hazards ratio estimated from nonparametric and Cox-based analyses. This plot allows for graphical testing of a proportional hazards assumption, which seems to hold in this case. Figure 6(b) shows that over the first 30 days after admission to an ICU, the RHC procedure reduces survival by 6.5% (95% CI is [4.1%, 8.9%]). This finding is consistent with the results of Connors et al.(1996) [3], who concluded that patients treated with RHC had, on average, lower survival probability, and they estimated the risk difference at 5.74% with 95% confidence interval of [3.40%, 8.09%]. Furthermore, using **causalCmprsk**, the risk difference measure can be translated into the domain of the expected number of days, lost as a result of having RHC, which is plotted on Figure 6(c). These results show that over the first 30 days after admission to an ICU, RHC reduces survival by 1.34 days (95% CI=[0.82, 1.84] days). The code that generated plots on Figure 6 and the results in Table 3 can be found in Supplementary Appendix available in the online version of this paper.

Additionally, the function `get.pointEst` can be used to extract point estimates corresponding to a time point of interest, for both treatment-specific parameters as well as for treatment effects. The following code illustrates how to obtain 30-day estimates of RD, RR, and difference in RMT, and their corresponding 95% confidence intervals.

```
> est30 <- get.pointEst(cmprsk.obj=res.30, timepoint=30)
> cat("Risk difference=", round((est30$trt.eff[[1]]$RD), dig=4),
+    ", 95% CI=[" , round((est30$trt.eff[[1]]$RD.CI.L), dig=4), " , " ,
+    round((est30$trt.eff[[1]]$RD.CI.U), dig=4), "]\n", sep="")
Risk difference=0.0652, 95% CI=[0.0413, 0.0885]
> cat("Risk ratio=", round((est30$trt.eff[[1]]$RR), dig=4),
+    ", 95% CI=[" , round((est30$trt.eff[[1]]$RR.CI.L), dig=4), " , " ,
+    round((est30$trt.eff[[1]]$RR.CI.U), dig=4), "]\n", sep="")
Risk ratio=1.2106, 95% CI=[1.1312, 1.2943]
> cat("RMT difference=", round((est30$trt.eff[[1]]$ATE.RMT), dig=4),
+    ", 95% CI=[" , round((est30$trt.eff[[1]]$ATE.RMT.CI.L), dig=4), " , " ,
+    round((est30$trt.eff[[1]]$ATE.RMT.CI.U), dig=4), "]\n", sep="")
RMT difference=1.3439, 95% CI=[0.8237, 1.8438]
```

5 Discussion

The package **causalCmprsk** provides a convenient and easy-to-use tool for causal analysis of competing risks data. Since its first release, **causalCmprsk** has been proven to be useful in several recently published biomedical papers from different research groups [1, 2, 48, 49]. We illustrate how to use **causalCmprsk** in a comprehensive analysis of the RHC data. As a caveat, we notice that we use the RHC data for illustration purposes and do not make any statements about clinical utility of the RHC procedure.

The package implements an IPW approach and provides a variety of weights: stabilized and non-stabilized weights for the overall, treated and control populations, and overlap weights for the overlap population, which is the population with the largest equipoise regarding the treatment. All the functionals can be estimated based on a nonparametric or a Cox-based approaches, and the proportional hazards assumption can be visually tested. The package provides several measures of treatment effects on time-to-event outcomes, among which the restricted-mean-time differences are the most appealing - they are able to provide an intuitive summary of treatment effects and quantify how much time on average was lost (or gained) due to treatment. In **causalCmprsk** we implemented the Bayesian bootstrap for quantification of uncertainty and incorporated an option of parallel computing to speed up bootstrap computations.

The **causalCmprsk** package is freely available for download from the Comprehensive R Archive Network (CRAN) website. All the results and figures presented in this paper can be reproduced using the code included in the Supplementary Appendix associated with this article, which can be found in the online version, and also in the vignette accompanying **causalCmprsk** at https://cran.r-project.org/package=causalCmprsk/vignettes/cmp_rsk_RHC.html.

One limitation of **causalCmprsk** is that the censoring distribution is assumed to be independent of covariates **C**, treatment assignment **A**, and potential outcomes (T^a, E^a) , which is often violated in practice. However, if the conditional independence between censoring time and T^a, E^a, A , given **C**, can be assumed, and if all the covariates are categorical, a stratified analysis by **C** can be carried out using **causalCmprsk**. Alternatively, an inverse probability of censoring approach can be applied to the selected uncensored observations. However, this strategy might be inefficient and result in the loss of information, if censoring proportion is high. Currently, **causalCmprsk** relies on the logistic regression for a propensity scores model and does not allow for time-varying covariates. Incorporation of alternative models for propensity scores and extension of **causalCmprsk** to time-varying covariates will be explored in future. Another limitation of **causalCmprsk** is that this package relies on the assumption of “no unmeasured confounding” (condition CA1 from Section 3.1). If it does not hold, which is often the case in practice, other newly developed methods have to be applied. One more limitation of **causalCmprsk** is that it does not allow for disentangling of a direct treatment effect on one time-to-event from its effect on a competing risk. Some newly developed methods, based on principal stratification ([50, 51]), can provide answers to the causal questions on direct and indirect treatment effects in semi-competing risks data. However, this would require follow-up after the occurrence

of a non-terminal event. Another limitation of **causalCmprsk** is the lack of simultaneous confidence bands. Finally, **causalCmprsk** is limited to binary interventions administered at one time point. If the interest is in estimating causal effects of treatments that are taken during a sustained period of time, a method for time-varying treatments has to be used (see, e.g., [52]).

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- Competing risks are omnipresent in clinical studies and have to be addressed during data analysis.
- **causalCmprsk** estimates average treatment effects of interventions on time-to-event outcomes with competing events.
- **causalCmprsk** uses propensity scores and inverse-probability-weighting to emulate baseline randomization in observational data.
- **causalCmprsk** allows for estimation of several time-varying measures of treatment effects on time-to-event outcomes.
- Restricted-mean-time differences quantify by how much time the treatment accelerates the recovery or postpones death.

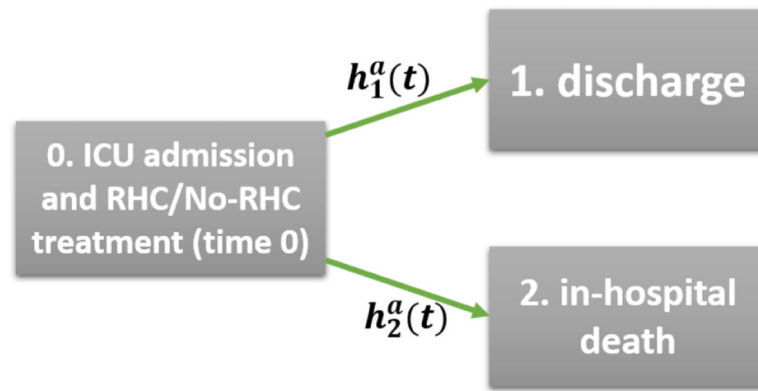


Figure 1:
A single-world (corresponding to treatment *a*) competing risks model with two competing events **1. Discharge** and **2. In-hospital death**.

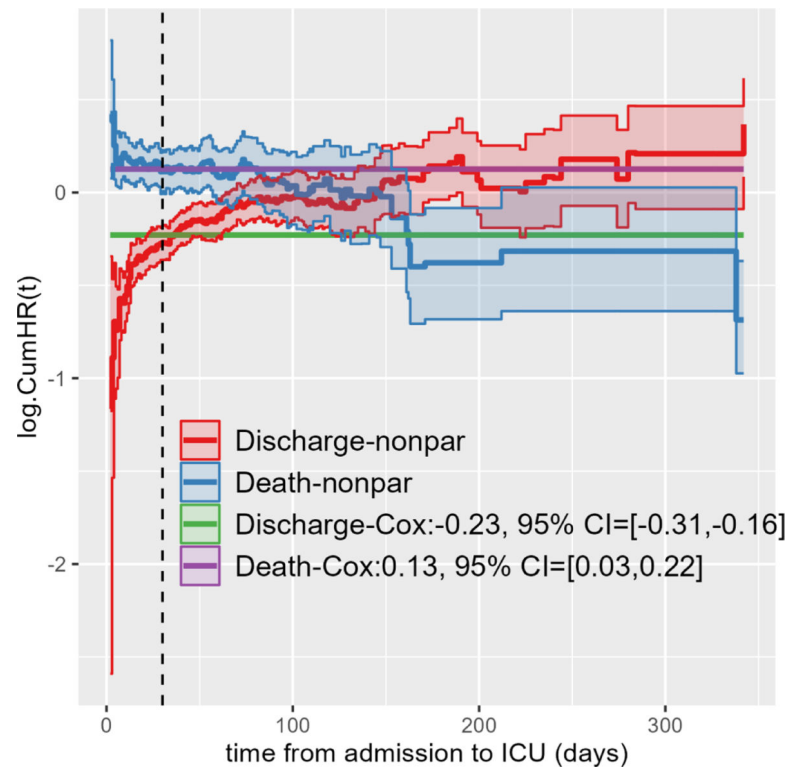


Figure 2:

A graphical test of a proportional hazards assumption. The plotted functions are

$\log \frac{H_k^{RHC}(t)}{H_k^{No-RHC}(t)}$ for transitions into the **Discharge** ($k = 1$) and **In-hospital death** ($k = 2$) states.

The vertical black dashed line marks 30 days past the admission to an ICU.

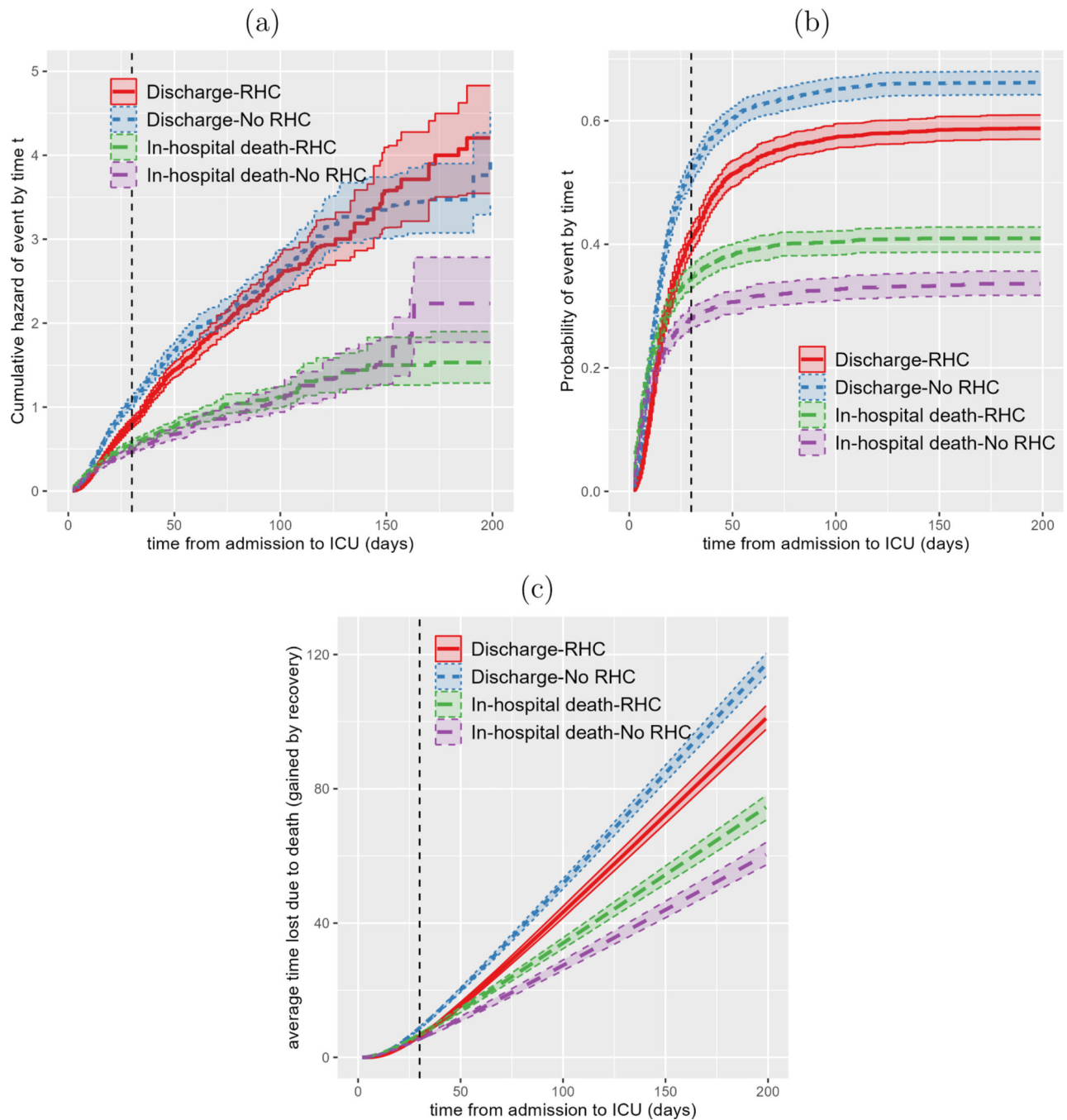


Figure 3:

(a) Cause-specific cumulative hazards; (b) Cumulative incidence functions; (c) Restricted-mean-time-lost/gained due to specific event. On all the panels, red solid lines represent **Discharge** outcome in the **RHC** group, blue dotted stand for **Discharge-No RHC**, green longdashed are for **In-hospital death-RHC**, and purple dashed are for **In-hospital death-No RHC**. The vertical black dashed line marks 30 days past the admission to an ICU.

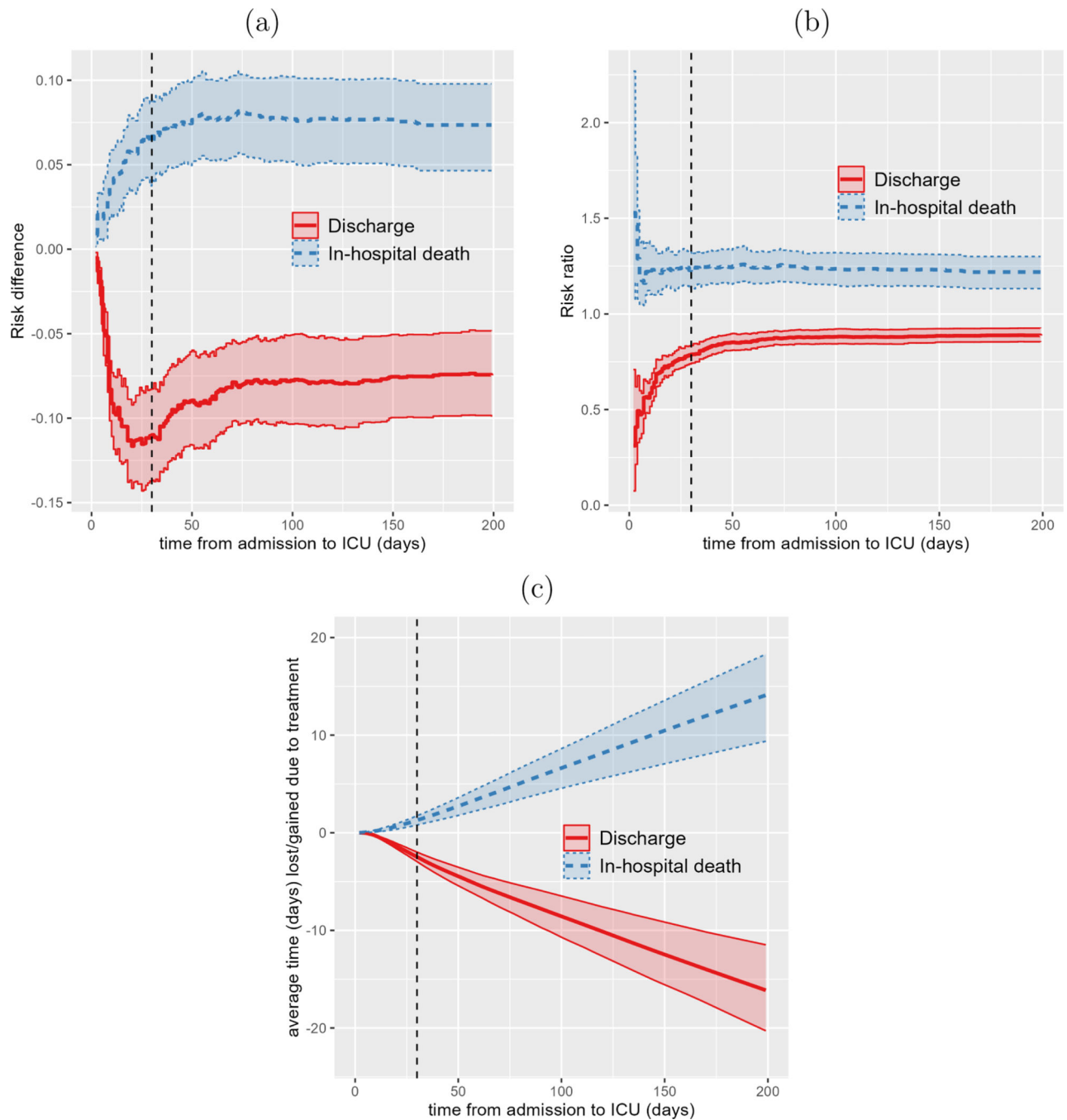


Figure 4:

(a) Risk differences for discharge and in-hospital death; (b) Risk ratios for discharge and in-hospital death; (c) Difference in average time lost/gained due to treatment. On all the panels, red solid lines represent treatment effects of RHC compared to No-RHC on the time to **Discharge**, and the blue dotted lines correspond to the treatment effects of RHC versus No-RHC on the time to **In-hospital death**. The vertical black dashed line marks 30 days past the admission to an ICU.

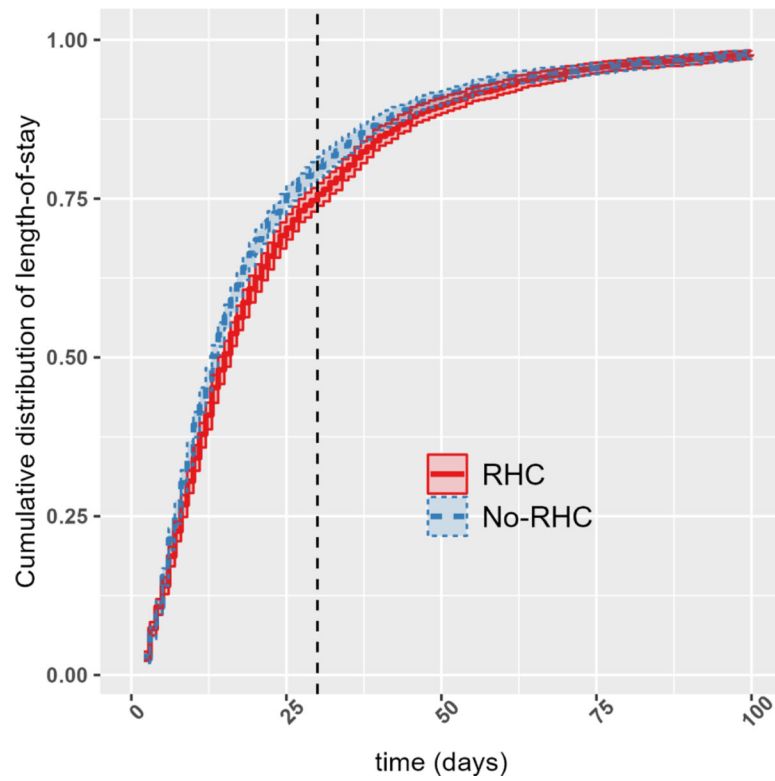


Figure 5:
Estimated single-world cumulative distribution functions of the length-of-stay in two treatment groups, i.e., $P(LOS^{RHC} \leq t)$ and $P(LOS^{noRHC} \leq t)$, with pointwise 95% confidence intervals. The vertical black dashed line marks 30 days past the admission to an ICU.

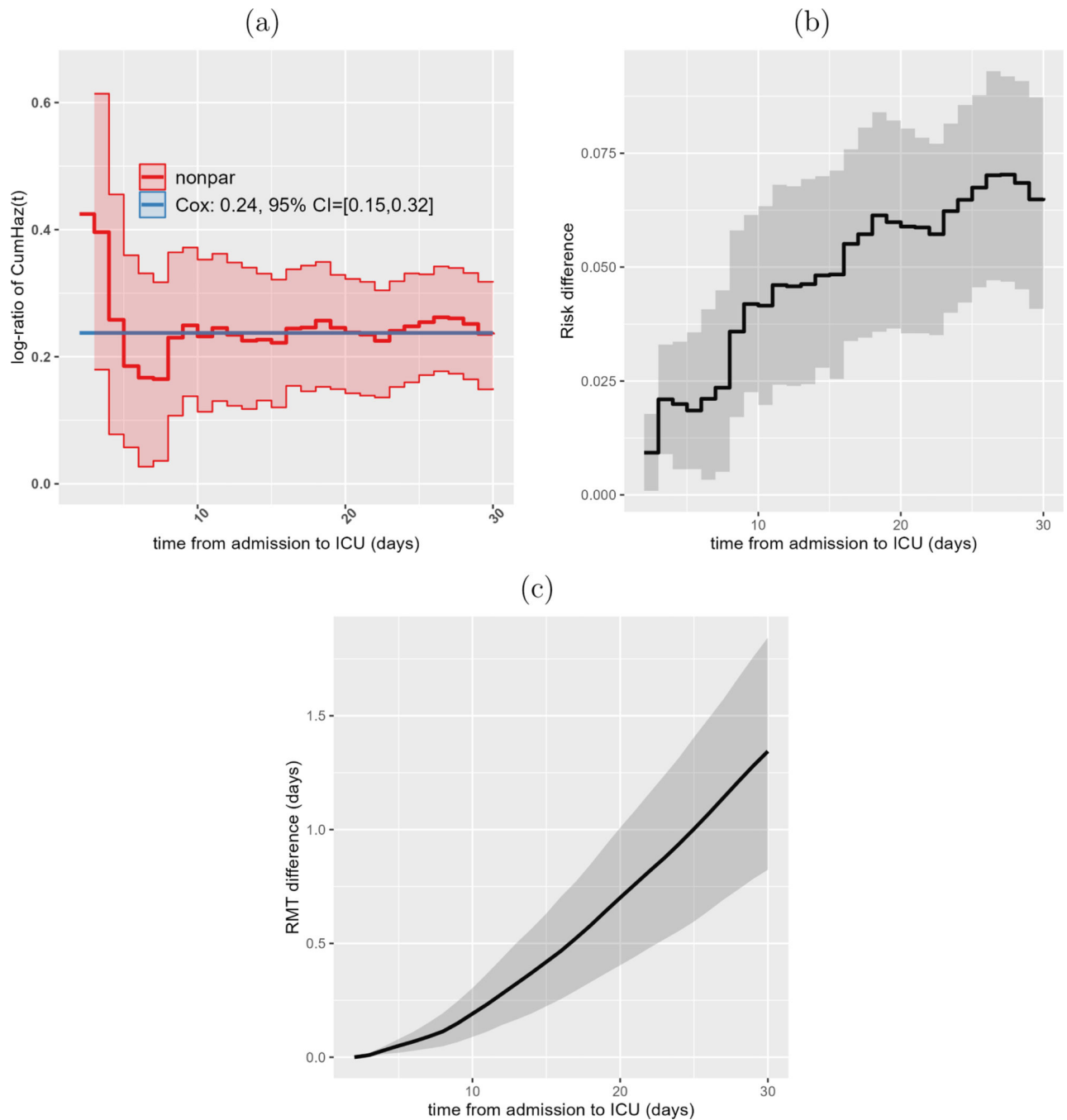


Figure 6:

Analysis of 30-day survival. (a) log-ratio of cumulative hazards of death comparing RHC-treated to non-treated patients. Under the Cox PH assumption, the $HR = \exp(0.24) = 1.27$ with 95% $CI = [1.16, 1.38]$; (b) Risk difference of death between RHC-treated vs non-treated patients; (c) Difference in restricted mean time, lost due to RHC treatment.

Table 1:
Input arguments of functions fit.cox and fit.nonpar

Arguments	Description
df	a data frame that includes time-to-event X, type of event E, a treatment indicator A and covariates C
X	a character string specifying the name of the time-to-event variable in df
E	a character string specifying the name of the “cause of failure” variable in df
trt.formula	a formula expression, of the form response ~ predictors. The response is a binary indicator of treatment/exposure, for which a logistic regression model (a Propensity Score model) will be fit using glm. As an alternative to specifying trt.formula, arguments A and C, defined below, can be specified. Either trt.formula or a pair of A and C must be specified.
A	a character specifying the name of the treatment variable It is assumed that A is a numeric binary indicator with 0/1 values, where A=1 denotes a treatment group, and A=0 denotes a control group.
C	a vector of character strings with variable names (of potential confounders).
wtype	a character string variable indicating the type of weights that will define the target population for which the ATE will be estimated. The possible values are “unadj”, “stab.ATE”, “ATE”, “ATT”, “ATC” and “overlap”.
cens	an integer value in E that corresponds to censoring times recorded in X. By default fit.nonpar assumes cens=0
conf.level	the confidence level that will be used in the bootstrap confidence intervals. The default is 0.95
bs	a logical value indicating whether to perform bootstrap in order to obtain confidence intervals
nbs.rep	number of bootstrap replications
seed	the random seed for the bootstrap, in order to make the results reproducible
parallel	a logical value indicating whether to perform in parallel using several cores simultaneously. The default value is FALSE.
verbose	a logical value indicating whether to show progress of bootstrap. The progress bar is shown only for sequential bootstrap computation.

Table 2:
Summary of results (point estimates with [95% Confidence Intervals]) of competing risks analyses in Section 4.2.

Parameter	$k = \text{Discharge}$		$k = \text{In-hospital death}$	
	$a = \text{RHC}$	$a = \text{No-RHC}$	$a = \text{RHC}$	$a = \text{No-RHC}$
$H_k^a(30)$	0.820 [0.763, 0.876]	1.073 [1.002, 1.136]	0.560 [0.519, 0.611]	0.491 [0.453, 0.538]
$CIF_a(30, k)$	0.410 [0.389, 0.429]	0.520 [0.498, 0.537]	0.347 [0.329, 0.367]	0.280 [0.263, 0.299]
$RMT_a(30, k)$	6.308 [5.921, 6.655]	8.780 [8.431, 9.116]	6.828 [6.406, 7.225]	5.544 [5.195, 5.950]
$\log HR_k(30)$	-0.270 [-0.363, -0.177]		0.132 [0.007, 0.228]	
$RD(30, k)$	-0.110 [-0.137, -0.083]		0.067 [0.041, 0.089]	
$RR(30, k)$	0.789 [0.743, 0.835]		1.239 [1.140, 1.330]	
$RMT_1(30, k) - RMT_0(30, k)$	-2.473 [-2.950, -1.976]		1.285 [0.783, 1.734]	

Table 3:
Summary (point estimates with [95% Confidence Intervals]) of analyses of univariate survival outcomes in Sections 4.3 and 4.4. LOS stands for the Length-Of-Stay in a hospital.

Parameter	<i>a</i> =RHC	<i>a</i> =No-RHC
Estimates for the Length-of-stay outcome		
$Pr_a(LOS \leq 30)$	0.757 [0.739, 0.774]	0.800 [0.783, 0.816]
$RMT_a^{LOS}(30)$	13.136 [12.703, 13.500]	14.324 [13.955, 14.663]
$Pr_1(LOS \leq 30) - Pr_0(LOS \leq 30)$	-0.043 [-0.066, -0.017]	
$RMT_1^{LOS}(30) - RMT_0^{LOS}(30)$	-1.188 [-1.7, -0.686]	
Estimates for the Time-to-death (in or out of hospital)		
$H^a(30)^*$	0.464 [0.437, 0.497]	0.367 [0.342, 0.394]
$CIF_a(30)^*$	0.375 [0.357, 0.396]	0.309 [0.292, 0.328]
$RMT_a(30)^*$	7.209 [6.826, 7.636]	5.865 [5.495, 6.231]
$\log HR(30)^*$	0.235 [0.150, 0.319]	
$RD(30)^*$	0.065 [0.041, 0.089]	
$RR(30)^*$	1.211 [1.131, 1.294]	
$RMT_1(30)^* - RMT_0(30)^*$	1.344 [0.824, 1.844]	

(*) The asterisk denotes the results of analyses of the time to death, which happened either in a hospital or after discharge.